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**Started on** Monday, 26 June 2023, 9:04 AM

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**State** Finished

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**Completed on** Monday, 26 June 2023, 9:05 AM

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**Time taken** 19 secs

Q10 Only

Question **1**

Not answered

Marked out of 14.00

\*Note that this question gives you the opportunity to upload a python notebook of your workings if necessary

1. Coordinate frame 1 is rotated by 45 degrees clockwise around the z axis and displaced by (2,1,1) with respect to coordinate frame 0. Provide answers correct to 3 decimal places.

1. Show the homogenous transformation matrix  $T_1^0$

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2. Point  $P^1 = [1,0,1]$ . What is  $P^0$ ?

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2. Coordinate frame 2 is rotated by 30 degrees anticlockwise around the x axis and displaced by (0,1,0) with respect to axis 1.

1. Calculate  $T_2^1$

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2. Calculate  $T_2^0$

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Question **2**

Not answered

Not graded

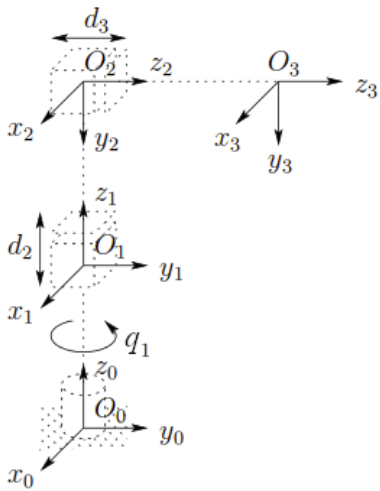
You can upload your python notebook here if you have one. Upload an empty file here if you want the question to be marked as complete but don't have anything to upload.

## Question 3

Not answered

Marked out of 6.00

The RPP arm used for this question is shown below. Note that the second joint is on top of the first one



Fill in the corresponding table of DH parameters (using degrees, not radians)

| $i$ | $a_i$                | $\alpha_i$           | $d_i$                | $\theta_i$           |
|-----|----------------------|----------------------|----------------------|----------------------|
| 1   | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| 2   | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| 3   | <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

## Question 4

Not answered

Marked out of 13.00

\*Note that you should show your workings in the provided answer book or upload your python notebook in the upload box below.

You have an RPR arm with the following transformation matrices.

Work out  $J_\omega$  and  $J_v$ .

$$T_1^0 = \begin{pmatrix} \cos(q1) & 0 & -\sin(q1) & 0 \\ \sin(q1) & 0 & \cos(q1) & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_2^0 = \begin{pmatrix} \cos(q1) & 0 & -\sin(q1) & -q2 \sin(q1) \\ \sin(q1) & 0 & \cos(q1) & q2 \cos(q1) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_3^0 = \begin{pmatrix} \cos(q1) \cos(q3) & -\sin(q1) & \cos(q1) \sin(q3) & -q2 \sin(q1) \\ \sin(q1) \cos(q3) & \cos(q1) & \sin(q1) \sin(q3) & q2 \cos(q1) \\ -\sin(q3) & 0 & \cos(q3) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Note that you should format terms as  $x\cos(y)$  where appropriate.

$J_\omega =$

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$J_v =$

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What is the linear velocity of the endpoint if joint 1 is rotated to  $\pi/6$  radians and is rotating at 1 radian per second, joint 2 is extended to 1 m and is not moving, and joint 3 is rotated to  $\pi/4$  radians and is rotating at 4 radians per second.

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Question **5**

Not answered

Not graded

You can upload your python notebook here if you have one. Otherwise we will mark the workings in your answer book. Upload an empty file here if you want the question to be marked as complete but don't have anything to upload.

Question **6**

Not answered

Marked out of 5.00

We talk of a robot as an embodiment of a perception-action loop.

For each of the following, is it best described as an example of sensing, planning, or acting?

Sending a signal to another robot with information of what it is planning on doing

Moving the arms of a humanoid to ensure balance

Feeling if the ground ahead of the robot is stable

Identifying regions of congested traffic on a road network

Predicting future stock prices

Determining the best way to divide up a search area between a team of mobile robots

Vocalising an instruction to a collaborating human

Identifying the dirtiest spots on a floor to be cleaned

Determining which crops are ready to be harvested

Choosing the best move to make in a game of chess

## Question 7

Not answered

Marked out of 5.00

Consider the following second-order system that we want to control:

$$2\ddot{x}_1 + 8\dot{x}_1 + u = 0 \text{ with } y = \dot{x}_1$$

By introducing a change of variables, rewrite this as a first-order system:

$$\dot{x} = Ax + Bu, \text{ where } \dot{x} = (\dot{x}_1, \dot{x}_2)^T$$

$A =$

(  ,  )

(  ,  )

$B^T =$

(  ,  )

Which of the following statements are true?

The system is linear:

The system is time-invariant:

The system is controllable:  because the rank of the controllability matrix is

The system is observable:  because the rank of the observability matrix is

## Question 8

Not answered

Marked out of 6.00

Consider the second-order system that we want to control:

$$\dot{x}_1 = -x_2$$

$$\dot{x}_2 = 2u$$

$$\text{With } x_1(0) = 0, x_2(0) = 0,$$

where  $x_1$  is the position of the system and  $u$  is the control variable, and the cost function is given by

$$J(u) = A(x_1(2) - 2)^2 + B(x_2(2))^2 + C \int_0^2 [u(t)]^2 dt$$

By introducing an appropriate Hamiltonian, find the control and state trajectory that minimises the cost function for  $A = 4, B = 1, C = 1$ .

From this, we have that  $x_1(t) =$    $t^3 +$    $t^2 +$    $t +$

Please note that the above values must be numerical.

Now assume we set  $A = 0, B = 2, C = 4$ . Which of the following statements are true?

We care more about the velocity than the position of the system: ☐

We would have an arbitrary final position: ☐

The final velocity should be close to 2: ☐

We care more about the integral cost than the terminal cost: ☐

## Question 9

Not answered

Marked out of 6.00

Consider the following matrix game, where the given payoffs are to the column player, who is the maximising player. The row player is minimising, and receives the negative of the shown reward:

$$\begin{pmatrix} 1 & -1 & 3 \\ 2 & 1 & -2 \\ 1 & -2 & 2 \end{pmatrix}.$$

The Nash equilibrium policy for the column player, where each number is the probability of picking that column, is given by

[  ,  ,  ]

The payoff of the game is .

Which of the following statements are true?

This game is cooperative: ☐

The game is shown here in normal form: ☐

This game can be turned into a three player game by considering a 3D table of payoffs: ☐

The Nash equilibrium is the highest payoff that any player can receive in the game: ☐

## Question 10

Not answered

Marked out of 5.00

A robotic spacecraft is hovering over an asteroid and needs to measure the distance  $x$  to the asteroid so as to extract a sample from it. The robot takes multiple readings to try and determine the distance before initialising the extraction procedure. Assume neither the spacecraft nor the asteroid moves after each reading, and so  $x_t = x_{t-1}$ . The noisy readings of the distance are given by  $z_t = x_t + \delta_t$ , where the noise is described by the process  $\delta_t \sim N(0; 0.1)$ .

Assume the current estimate is  $x_0 = 0.35m$ , and  $P_0 = 1$ . Now the spacecraft receives a reading of  $z_1 = 0.49m$ . Using a Kalman filter, obtain the updated mean and variance of the spacecraft's estimate to the asteroid.

After the update  $x_1 =$   and  $P_1 =$  .

The Kalman gain  $K =$  .

Which of the following statements are true about Kalman filters?

The value of  $x$  in the predictor step  $x_t^-$  is always greater than the value of  $x$  in the corrector step  $x_t$ :

A Kalman filter can only be directly applied to a problem if the system [dynamics](#) are known:

If the distribution over  $x$  is a Gaussian, then a Kalman filter will give a more accurate estimation than a Bayes filter:

A particle filter is a Bayes filter where we use particles to approximate the mean and variance of the distribution:

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